## 1.13 Betragsgleichungen

(a) $|2-5x| = 3$

$$\Rightarrow \quad 2-5x = +3 \quad \vee \quad 2-5x = -3$$
$$\Rightarrow \quad -5x = 1 \quad \vee \quad -5x = -5$$
$$\Rightarrow \quad x = -1/5 \quad \vee \quad x = 1$$

(b) $|x^2-9| = 7$

$$\Rightarrow \quad x^2-9 = +7 \quad \vee \quad x^2-9 = -7$$
$$\Rightarrow \quad x^2 = 16 \quad \vee \quad x^2 = 2$$
$$\Rightarrow \quad x = 4 \vee x = -4 \vee x = +\sqrt{2} \vee x = -\sqrt{2}$$